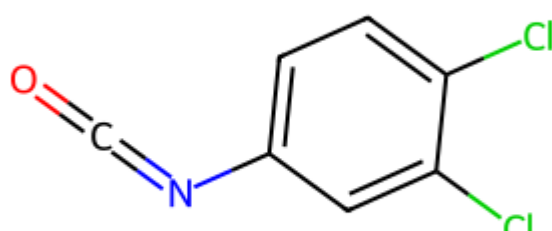


```
{'analysis_time': 0.0,  
'first_solution_time': 0.264314,  
'num_calls_rxn_model': 0,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 3.493181100278343,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 3.493181100278343,  
'retro_star_value': 3.493181100278343}
```

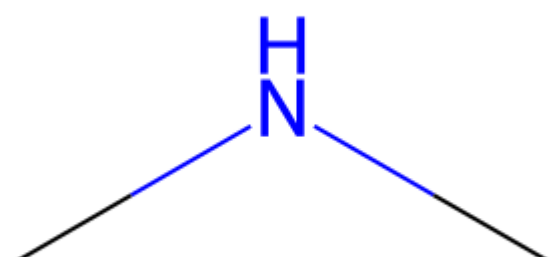


Reaction

```
{'analysis_time': 0.169289,  
'first_solution_time': 0.264314,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 3.493181100278343,  
'retro_star_reaction_number': 3.493181100278343,  
'retro_star_rxn_cost': 0.36961545521446726,  
'retro_star_value': 3.493181100278343}
```



```
{'analysis_time': 0.169289,  
'first_solution_time': 0.169289,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 3.493181100278343}
```

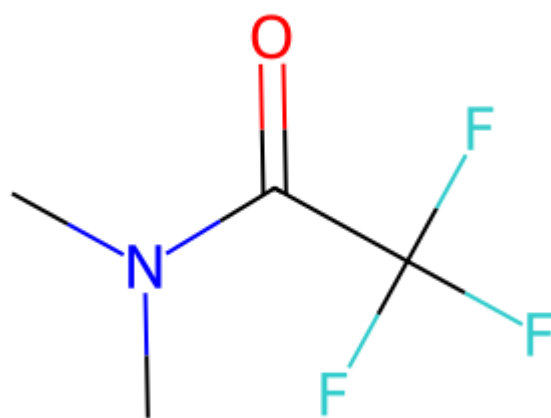


```
{'analysis_time': 0.169289,  
'first_solution_time': 0.264314,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 3.123565645063876,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 3.123565645063876,  
'retro_star_value': 3.493181100278343}
```



Reaction

```
{'analysis_time': 0.264314,  
'first_solution_time': 0.264314,  
'num_calls_rxn_model': 2,  
'num_calls_value_function': 50,  
'retro_star_min_cost': 6.214608098422191,  
'retro_star_reaction_number': 6.214608098422191,  
'retro_star_rxn_cost': 6.214608098422191,  
'retro_star_value': 6.584223553636658}
```



```
{'analysis_time': 0.264314,  
'first_solution_time': 0.264314,  
'num_calls_rxn_model': 2,  
'num_calls_value_function': 50,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 6.584223553636658}
```